ai-construct Command Computer Spec

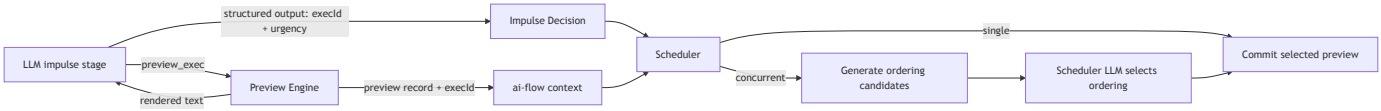
> v2 — Preview/Commit model. Supersedes v1 (exec-only model). See `command-computer-overview.md` for the usability-focused summary.

---

1. Overview

An ai-construct is an AI persona backed by a **computer** — a persistent Linux-like environment the AI can read, write, run code on, and accumulate capability in over time. Every impulse runs on the same computer.

The AI has **one tool:** `preview_exec`. It writes TypeScript that runs on a **forked** world — nothing commits. The AI can call `preview_exec` multiple times to explore, test, and draft. Each preview produces an `execId`. The impulse stage's structured output selects one `execId` to commit. The scheduler later commits the selected preview against real state.



The system has three layers:

```
ai-flow              generic LLM primitives (createPreviewExecTool, OverlayFs)
ai-sandbox-computer  the computer (command registry, preview engine, sys modules)
ai-construct         the persona (impulse lifecycle, scheduler, intent hints,
nap/hypno)
```

ai-construct calls into ai-flow. ai-flow has no knowledge of ai-construct.

---

2. Terminology

Term	Definition
`preview_exec`	The AI's single tool. Runs TypeScript on a forked world. Returns rendered text to the LLM + stores a preview record in ai-flow context.
preview record	The full artifact from a `preview_exec` call: `execId`, code, stdout, draft responses. Stored hidden from the LLM in ai-flow flow context.

Term	Definition
`execId`	Stable identifier for a preview record. The LLM selects one in structured output.
selected preview	The preview record chosen by the impulse structured output's `execId`. The scheduler commits this.
command	A registered operation (read, write, list, etc.) invoked via the `command()` function.
command handler	The function that implements a command. Registered via `defineCommand`.
command registry	The map of command names → handlers. Three tiers: built-in, developer, AI-installed.
code binding	A typed product API function imported as an ESM module. Defined via `defineCodeBinding` / `.tool.ts`.
`show()`	Global that pretty-prints a value (using the command's `render`) and returns the raw value.
`draft_response()`	Global that proposes a user-facing response. Not sent immediately — the scheduler decides delivery.
intent hint	A TOML entry mapping a natural-language phrase to a concrete command invocation. Auto-previewed at impulse start.
prompt window	Tokens the LLM is actively reasoning over.
flow state	Typed TypeScript object ai-flow maintains across stages.
host context	Server-side `{ deps, fs, ctx, input }` passed to CodeFunction implementations. `deps` are invocation-scoped and may be tx-bound.
transaction cycle	One top-level request / invocation lifecycle that opens a DB transaction, binds invocation-scoped deps, runs handlers, then commits or rolls back.
effect context	Invocation-scoped fire-and-forget effect collector. Handlers call effects immediately for preview text, but real delivery only happens after commit succeeds.
construct memory	Everything persisted in the VFS.
impulse payload	Structured or unstructured data that triggered an impulse.

---

3. Package Boundaries

`@inkibra/ai-flow`

Generic LLM pipeline primitives. No knowledge of constructs, computers, commands, or intents.

Owns:

- `createAiFlow`, `createAiOutputStage`, `createAiTextStage` — flow builder
- `createPreviewExecTool` — the single AI tool factory
- **Two tool kinds:** JSON-schema function tools (existing) and **text-input custom tools** (new — required for `preview_exec`). OpenAI Responses API supports custom tools with free-form text inputs and outputs. ai-flow needs to expose this alongside the existing function-tool path.
- Hidden preview record storage in flow context (per-stage `previewExecRuns` map)
- `OverlayFs`, `createOverlayFs` — VFS layer
- `CodeFunction`, `codeFunction()`, `defineCodeBinding` — product binding types
- `.tool.ts` build-pack plugin — generates validators and declarations
- OpenAI streaming orchestration (`flow.ts`)

Does not own: command registry, defineCommand, sys modules, preview engine, ts-pm, router transaction cycles, effect contexts, intent hints, impulse lifecycle, scheduler.

`@inkibra/ai-sandbox-computer` *(new package)*

The computer abstraction. Can be used independently of ai-construct.

Owns:

- `CommandRegistry` — the command map (built-in + developer + AI-installed)
- `defineCommand` — the universal command definition interface
- All built-in command handlers (read, write, list, search, open, cron, pm, etc.)
- Preview engine — `vm.SourceTextModule` on a forked OverlayFs plus a router transaction cycle for DB-backed product APIs
- `sys` SyntheticModule — exports `command()` for use inside preview code
- `sys/ai` SyntheticModule — lightweight LLM SDK (extract, search, generate)
- Node built-in proxy SyntheticModules (node:fs, node:path, etc.)
- `show()` global — pretty-print via command's `render`, return raw value
- `draft_response()` global — propose user-facing response
- `createComputer(overlayFs, config)` — assembles registry + engine
- `defineAiComputerModule()` — groups related code bindings + handler factories
- Router execute-context mapping callback for AI invocations

Does not own: HTTP backend request lifecycle, route definitions, context codecs, workflow runtime internals, impulse lifecycle, intent hints, scheduler.

`@inkibra/router`

Typed routes, handler/provider abstraction, context codecs, and the shared transaction-cycle interface used by both HTTP backends and ai-computer.

Owns:

- `createAPIRoute`, `defineRouteSchema`, `defineContextSchema`
- `createApiRouteHandler` and provider-friendly handler typing
- Transaction-cycle interface for one request / invocation lifecycle
- Side-effect interface / effect context contract
- Route metadata for preview-safe vs preview-unsafe exposure

Does not own: concrete DB driver implementation, DAL collection factories, workflow persistence, construct scheduling.

`@inkibra/dal-connection`

Database driver layer. Implements the router transaction-cycle interface for DB-backed runtimes.

Owns:

- `Driver` and transaction primitives
- Transaction-cycle implementation for supported drivers
- Tx-bound driver wrapper used to instantiate DAL collections per invocation

Does not own: route definitions, effect dispatch semantics, construct logic.

`@inkibra/workflow`

Durable deferred work. Used as an effect target after commit, not as inline preview-time external execution.

Owns:

- Workflow runtime / scheduler / queue system
- Durable workflow start APIs
- Retry / resume semantics for external work

Does not own: request transaction lifecycle, inline handler side effects.

`@inkibra/ai-construct`

The persona layer. Uses ai-sandbox-computer as its computer. Calls into ai-flow for LLM reasoning.

Owns:

- Impulse lifecycle (`runner.ts`, `flow.ts`, `construct.ts`)
- Intent hint registry (TOML), similarity matching, auto-preview of hints
- Impulse structured output schema (`thinking`, `urgency`, `execId`)

- Scheduler: commit of selected previews, ordering candidate generation, concurrent ordering selection
- Nap and hypno flows
- Response delivery based on `draft_response()` proposals + `urgency`
- Per-impulse auth/context assembly passed into ai-computer execute-context mapping

---

4. VFS Layout

The construct's filesystem follows Linux conventions. The AI uses standard paths. Path semantics are enforced by OverlayFs permission layers.

```
/
├── usr/
│   ├── lib/
│   │   └── node_modules/     system modules (product bindings + bundled npm)
│   │       ├── workout-api/
│   │       │   ├── index.d.ts   AI reads this for type discovery
│   │       │   └── package.json
│   │       ├── zod/             pre-installed (esbuild bundle)
│   │       └── date-fns/        pre-installed (esbuild bundle)
│   └── share/
│       └── construct/
│           └── capabilities.d.ts   type declarations for system modules
├── home/
│   └── user/
│       ├── context/          AI's working memory (documents, state, notes)
│       │   └── state/        well-known state files (scheduler reads these)
│       ├── scripts/          AI-authored scripts and in-progress packages
│       └── .local/
│           ├── lib/
│           │   ├── node_modules/  agent-authored ts-pm packages
│           │   │   └── @commands/ AI-installed commands
│           │   │       └── streak-check/
│           │   └── intents/
│           │       └── registry.toml   intent hint registry (see §18)
│           └── bin/          AI-created executable scripts
├── var/
│   ├── lib/
│   │   └── ts-pm/
│   │       └── registry/     immutable published ts-pm packages
```

```
│   │           └── <pkg>/<version>/
│   ├── log/                  structured logs
│   └── spool/
│       └── cron/             scheduled impulses and reminders
├── etc/                      system configuration (AI read-only)
│   └── construct.md          construct identity and instructions
├── proc/
│   ├── self/
│   │   └── ctx               current flow state (readable by AI)
│   ├── meminfo               construct memory stats (file count, bytes, dirty)
│   ├── contextinfo           context window token budget
│   └── loadavg               active impulse count
└── tmp/                      ephemeral scratch space (cleared between naps)
```

Write permissions:

- AI can write anywhere under `/home/user/`
- AI can write to `/var/spool/cron/` (cron entries), `/var/log/` (append only)
- `/usr/`, `/etc/`, `/proc/` are read-only to AI
- `/tmp/` is writable but ephemeral

---

5. The `preview_exec` Tool

5.1 Schema

The AI's single tool during the impulse stage:

```
{
  name: 'preview_exec',
  type: 'custom',            // text-input tool, not JSON-schema function tool
  description: '...',        // auto-generated from command registry + bindings
}
```

`preview_exec` accepts a **TypeScript program as raw text** — not a JSON object with a `code` field. The model writes code directly as the tool input.

This requires ai-flow to support **custom / text-input tools** alongside the existing JSON-schema function tools. The OpenAI Responses API supports custom tools with free-form text inputs and outputs. If a specific provider does not support custom tools, ai-flow may fall back to wrapping the input as `{ code: string }` — but this is a transport detail, not the conceptual API.

One tool. Raw text input. **Nothing commits.**

5.2 What Happens on Each Call

1. Code is transpiled (TypeScript → JS via `Bun.Transpiler`)
2. A top-level **transaction cycle** is opened for DB-backed product APIs
3. Code runs on a **forked OverlayFs** — VFS writes are visible within the preview but do not affect real state
4. Code bindings call router-style handlers/providers using invocation-scoped deps built from the tx-bound driver and effect context
5. All `command()` calls work normally against the forked VFS
6. `show()` and `console.log` output is captured as stdout
7. `draft_response()` proposals are captured in the preview record
8. Effect calls return preview text immediately, but are only staged in the effect context
9. The preview engine produces a `PreviewExecRecord` with an `execId`
10. The transaction cycle always rolls back at the end of the preview

5.3 Two Consumers

`preview_exec` serves two audiences simultaneously:

The LLM receives rendered text to continue reasoning:

```
Current streak: 12 days
Preview: log-workout would succeed
Draft response: "Logged your squats! Streak: 13 days."

[exec_id: prev_01HXYZ]
```

ai-flow context receives the full preview record, stored hidden from the LLM in `ctx.previewExecRuns[execId]`.

This split uses ai-flow's existing tool architecture: `render()` produces text for the model, `execute()` mutates flow context with the full record.

5.4 Tool Description

The tool description sent to the LLM includes:

- List of available commands with signatures and descriptions (auto-generated from the command registry)
- List of available product binding modules with their declarations
- Note on `show()` / `draft_response()` usage
- Note on `sys/ai` availability and budget
- Reminder that previews do not commit — the structured output selects one

5.5 `createPreviewExecTool`

```
// ai-flow — uses the new text-input / custom tool kind
export function createPreviewExecTool<TFlowCtx extends Record<string, unknown>,
TDeps>(
  options: PreviewExecToolOptions,
): AiCustomTool<TFlowCtx, string, PreviewExecRecord, TDeps> {
  return createAiCustomTool('preview_exec', {
    description: generatePreviewExecDescription(options),
    parseInput: (raw: string) => ({ success: true, data: raw }),
    execute: async (code, ctx, deps) => {
      const record = await options.engine.preview(code);
      // Store in hidden flow context
      ctx.previewExecRuns = ctx.previewExecRuns ?? {};
      ctx.previewExecRuns[record.execId] = record;
      ctx.previewExecOrder = ctx.previewExecOrder ?? [];
      ctx.previewExecOrder.push(record.execId);
      return { success: true, data: record };
    },
    render: (record) => {
      const parts = [];
      if (record.stdout) parts.push(record.stdout);
      if (record.draftResponses.length > 0) {
        parts.push('Draft responses:');
        for (const r of record.draftResponses) {
          parts.push(`  [${r.importance ?? 'normal'}] ${r.text}`);
        }
      }
      parts.push(`\n[exec_id: ${record.execId}]`);
      return parts.join('\n');
    },
  });
}
```

`createAiCustomTool` is the new ai-flow factory for text-input tools. It differs from `createAiTool` in that it has `parseInput(raw: string)` instead of `parameterSchema` + `parseParameters`. ai-flow serializes it as a custom tool in the provider request, falling back to a single-field function tool if the provider does not support custom tools.

---

6. Preview Record Model

6.1 Shape

```
type PreviewExecRecord = {
  execId: string;                    // stable identifier, e.g. 'prev_01HXYZ...'
  code: string;                      // the TypeScript source that was previewed
  stdout: string;                    // captured console.log / show() output
  draftResponses: DraftResponse[];   // captured draft_response() calls
  effectPreviews: string[];          // preview text returned by staged effects
  createdAt: string;                 // ISO timestamp
};

type DraftResponse = {
  text: string;
  importance?: 'low' | 'normal' | 'high' | 'urgent';
};
```

6.2 Storage in ai-flow Context

Preview records are stored in the flow context, hidden from the LLM:

```
type ImpulseFlowContext = {
  impulse: Impulse;
  decision: ImpulseDecisionOutput | null;
  previewExecRuns: Record<string, PreviewExecRecord>;
  previewExecOrder: string[];   // insertion order for debugging
};
```

The LLM only sees the rendered text from `preview_exec`. The full records are available to ai-construct after the output stage completes, for passing to the scheduler.

6.3 Multiple Previews

The AI can call `preview_exec` multiple times in one impulse. Each call gets its own `execId`. The AI explores, tests, iterates — then picks the best one. Only the selected preview is committed.

---

7. Impulse Structured Output

7.1 Schema

The impulse stage uses `createAiOutputStage` with:

```
type ImpulseDecisionOutput = {
  thinking: string;
  urgency: 'none' | 'defer' | 'low' | 'normal' | 'urgent' | 'now';
  execId: string | null;
};
```

- **`thinking`**: Short internal note (max ~280 chars). Used for nap analysis.
- **`urgency`**: Stage-level routing hint for the scheduler / response stage.
- **`execId`**: References a preview record from `ctx.previewExecRuns`. If `null`, no program is committed (the impulse was observation-only).

7.2 Validation

After the output stage, ai-construct validates:

- If `execId` is non-null, it must exist in `ctx.previewExecRuns`
- If not found, the impulse fails with a clear error

7.3 What Gets Passed to the Scheduler

```
{
  impulseId: string;
  thinking: string;
  urgency: string;
  selectedPreview: PreviewExecRecord | null;  // resolved from execId
}
```

---

8. Command Registry & `defineCommand`

8.1 `defineCommand` Interface

```
type ArgDef = {
  type: 'string' | 'number' | 'boolean';
```

```
  position?: number;         // positional arg index (0-based)
  flag?: string;             // e.g. '--head', '--format'
  required?: boolean;        // default false
  default?: string | number | boolean;
  description?: string;
};

type CommandContext = {
  fs: OverlayFs;
  registry: CommandRegistry;  // call other commands
  deps?: unknown;             // developer-provided deps (for developer commands)
};

type Command<TResult = unknown> = {
  name: string;
  description: string;
  args: Record<string, ArgDef>;
  fn: (parsed: Record<string, unknown>, ctx: CommandContext) => Promise<TResult>;
  render: (result: TResult) => string;
};

function defineCommand<TResult>(opts: {
  name: string;
  description: string;
  args: Record<string, ArgDef>;
  fn: (parsed: Record<string, unknown>, ctx: CommandContext) => Promise<TResult>;
  render: (result: TResult) => string;
}): Command<TResult>;
```

Arg parsing: A lightweight runtime parser handles positional args and flags. `command('read', '/home/user/notes.md', '--head', '10')` parses to `{ path: '/home/user/notes.md', head: 10 }`. The parser validates types (is `--head` a number?) and required fields.

Auto-help: `command('read', '--help')` generates help text from the command's `description` and `args` definitions.

8.2 Built-in Commands

All built-in commands are registered via `defineCommand`.

File Operations

Command	Args	Returns	Description
`read`	`path` (pos 0, required), `--head N`, `--tail N`	`string` (file content)	Read a file, optionally first/last N lines
`write`	`path` (pos 0, required), `content` (pos 1, required)	`{ path, bytes }`	Create or overwrite a file
`edit`	`path` (pos 0, required), `old` (pos 1, required), `new` (pos 2, required), `--count N`	`{ replacements }`	Targeted string replacement
`list`	`path` (pos 0, default `.`)	`FsEntry[]`	List directory contents
`search`	`pattern` (pos 0, required), `path` (pos 1, default `.`), `--recursive`	`SearchMatch[]`	Search file contents
`find`	`path` (pos 0, required), `--name pattern`	`string[]` (paths)	Find files by name pattern
`stat`	`path` (pos 0, required)	`FileStat`	File metadata (size, lines, words, modified)
`remove`	`path` (pos 0, required)	`{ removed }`	Delete a file
`mkdir`	`path` (pos 0, required)	`{ created }`	Create directory (recursive)

Important: `show(await command('read', ...))` uses the `read` command's `render` function (e.g., line-numbered output). `console.log(await command('read', ...))` prints the raw string. Commands should always define a useful `render`.

Context Management

Command	Args	Returns	Description
`open`	`path` (pos 0, required)	file/dir content	Add file/dir to persistent context window. Supports globs: `dir/*`, `dir/**`. LRU+priority scoring, TTLs.
`close`	`path` (pos 0, required)	`{ closed }`	Remove from persistent context
`opened`	*(none)*	`OpenedEntry[]`	List open files with token counts
`pin`	`path` (pos 0, required), `--phase awake|nap|both`, `--reason text`, `--mode full|frontmatter`	`{ pinned }`	Pin in context window
`unpin`	`path` (pos 0, required)	`{ unpinned }`	Unpin from context

Command	Args	Returns	Description
`status`	`--phase awake|nap`	`ContextStatus`	Context pressure view (tokens used, budget, pins)

Scheduler

Command	Sub-command	Args	Description
`cron`	`list`	*(none)*	List all reminders
`cron`	`show`	`name` (pos 1)	Show reminder details
`cron`	`set`	`name` (pos 1), `--at time` or `--every cron-expr`, `--text msg`, `--timezone tz`	Create/update reminder
`cron`	`snooze`	`name` (pos 1), `--for duration`	Snooze a reminder
`cron`	`done`	`name` (pos 1)	Mark reminder done
`cron`	`rm`	`name` (pos 1)	Delete reminder

Package Management

Command	Sub-command	Args	Description
`pm`	`init`	`name` (pos 1)	Scaffold package in `/home/user/scripts/<name>/`
`pm`	`publish`	`name` (pos 1)	Compile TS→JS, copy to registry (immutable)
`pm`	`add`	`name` (pos 1), `--version ver`	Install. Auto-registers command if default export is a `Command`.
`pm`	`list`	*(none)*	List installed packages with lock status
`pm`	`versions`	`name` (pos 1)	List published versions
`pm`	`remove`	`name` (pos 1)	Uninstall, unregister command
`pm`	`link`	`path` (pos 1)	Dev-mode symlink to agent node_modules

8.3 Three Tiers

Tier	Registered by	Runs on	Example
Built-in	ai-sandbox-computer at init	Host (direct OverlayFs calls)	`read`, `write`, `list`, `open`, `cron`, `pm`
Developer	Product at construct init via config	Host (full Bun access, deps available)	`deploy`, `sync-data`
AI-installed	Auto-registered when `pm add` installs a package whose default export is a `Command`	Preview engine (vm.SourceTextModule)	`@commands/streak-check`

8.4 The `command()` Global

`command` is a pre-loaded global in the preview exec context:

```
async function command(name: string, ...args: (string | number | boolean)[]):
Promise<unknown>
```

- First arg is the command name
- Remaining args are passed to the arg parser
- Returns the command handler's result (structured, typed per command)
- For `write` and `edit`, the last arg is the content body
- Throws with stderr if the command fails

Also available via import: `import { command } from 'sys';`

---

9. `show()` and Output Handling

9.1 The `show()` Global

```
function show<T>(value: T): T
```

Pretty-prints a value for the LLM (writes to stdout → tool result) and returns the raw value for continued use in code.

How it formats: If the value came from a `command()` call, `show` uses the command's `render` function. Otherwise it falls back to:

- `string` → printed as-is
- `object` / `array` → `JSON.stringify(value, null, 2)`

- primitives → `String(value)`

Implementation: The `command()` function tags its return value with the command name (via a non-enumerable symbol property). `show()` checks for this tag and looks up the command's `render` function in the registry.

```
const COMMAND_TAG = Symbol('command');

async function command(name, ...args) {
  const result = await registry.dispatch(name, args);
  if (result && typeof result === 'object') {
    Object.defineProperty(result, COMMAND_TAG, { value: name, enumerable: false
});
  }
  return result;
}

function show(value) {
  const tag = value?.[COMMAND_TAG];
  if (tag) {
    console.log(registry.get(tag).render(value));
  } else if (typeof value === 'string') {
    console.log(value);
  } else {
    console.log(JSON.stringify(value, null, 2));
  }
  return value;
}
```

9.2 Stdout Capture

The preview engine captures all `console.log` output. This becomes part of the `PreviewExecRecord.stdout` and is rendered back to the LLM as the tool result. The AI does not need to `console.log` everything — only what it wants to see.

---

10. `draft_response()`

10.1 Shape

```
async function draft_response(response: {
  text: string;
  importance?: 'low' | 'normal' | 'high' | 'urgent';
}): Promise<void>
```

10.2 Semantics

- **Does not send immediately.** It appends a response proposal to the preview record's `draftResponses` array.
- The scheduler / response stage decides whether and when to deliver.
- Multiple `draft_response()` calls are allowed per preview.
- `importance` is a per-message delivery hint (distinct from the impulse-level `urgency` in structured output).

10.3 Why `draft_response()` in Code

The AI's response can be conditional on computed results:

```
const streak = JSON.parse(await command('read',
'/home/user/context/state/streak.json'));
await command('write', '/home/user/context/state/streak.json',
  JSON.stringify({ ...streak, count: streak.count + 1 }));

if (streak.count + 1 >= 30) {
  await draft_response({
    text: `Logged your squats! 🔥 30-day streak — incredible!`,
    importance: 'high',
  });
} else {
  await draft_response({
    text: `Got it, squats logged. Streak: ${streak.count + 1} days.`,
    importance: 'normal',
  });
}
```

The response is written after mutations, where the AI knows the actual outcome.

10.4 Relationship to Structured Output `urgency`

Concern	Source	Decides
Scheduling / routing	`urgency` in impulse structured output	When / whether to deliver
Per-message delivery hint	`importance` in `draft_response()`	Priority among multiple messages

Both are available to the scheduler. The AI declares urgency from its high-level assessment; importance is set programmatically per response.

---

11. Execution Engine

11.1 Architecture

```
preview_exec tool handler
  └─ engine.preview(code)
       ├─ Bun.Transpiler({ loader: 'ts' }).transformSync(code) → JS
       ├─ overlayFs.fork() → forked VFS
       ├─ router transaction cycle → tx-bound driver + effect context
        └─ vm.SourceTextModule(js, { context: vmContext })
            ├─ linker: 'sys'         → SyntheticModule { command }
            ├─ linker: 'sys/ai'      → SyntheticModule { ai }
            ├─ linker: 'workout-api' → SyntheticModule { getWorkouts, ... }
           ├─ linker: 'node:fs'     → SyntheticModule (forked OverlayFs proxy)
           ├─ linker: 'node:path'   → SyntheticModule (pure JS)
           ├─ linker: 'zod'         → SourceTextModule (loaded from VFS)
           └─ linker: '@commands/*' → SourceTextModule (loaded from VFS)
```

Everything on the same thread. SyntheticModules call host functions directly. No Workers, no SharedArrayBuffer, no bridge protocol. The forked OverlayFs ensures preview VFS writes don't affect real state. The transaction cycle ensures preview DB writes don't affect real state.

11.2 Preview vs Commit

The preview engine has two modes:

Mode	VFS	DB-backed bindings	Effects	Used by
Preview	Forked OverlayFs	Run inside a tx-bound invocation, then rollback	Staged only, preview text captured	`preview_exec` tool

Mode	VFS	DB-backed bindings	Effects	Used by
Commit	Real OverlayFs	Run inside a fresh tx-bound invocation, then commit	Flushed only after successful commit	Scheduler commit phase

The commit phase re-runs the exact code from the selected preview record, but against the real OverlayFs and a fresh transaction cycle. The selected preview is still the commit unit; DB persistence is controlled by transaction commit rather than by `isDryRun` branches or query-cache replay.

DB preview semantics: Read-after-write consistency comes from the open DB transaction itself. VFS preview semantics still come from `OverlayFs.fork()`. `@inkibra/query-cache` may still be used as a local performance helper, but it is no longer the correctness mechanism for DB-backed preview state.

11.2a Concrete Router Transaction / Effect API

The transaction cycle should be defined at the `@inkibra/router` level and implemented by `@inkibra/dal-connection` for supported drivers. Server runtimes and ai-computer both consume the same contract.

```
type TransactionCycleMode = 'http' | 'preview' | 'commit';

type EffectIntent = {
  kind: string;
  payload: unknown;
  preview?: string;
};

type EffectPreview = {
  text: string;
};

type EffectContext = {
  call(intent: EffectIntent): Promise<void>;
  getPreviews(): EffectPreview[];
};

type TransactionCycle = {
  driver: Driver;              // tx-bound driver for DAL instantiation
  effects: EffectContext;      // invocation-scoped staged effects
  commit(): Promise<void>;     // commit tx, then allow effect flush
  rollback(): Promise<void>;   // rollback tx, discard staged effects
};
```

```
type TransactionRuntime = {
  begin(args: {
    mode: TransactionCycleMode;
  }): Promise<TransactionCycle>;
};
```

Effect semantics:

- `effects.call(...)` is **fire-and-forget only**. It never returns business data to the handler.
- During preview, `effects.call(...)` only records preview text and staged intents; no real side effect runs.
- During commit / HTTP, intents remain staged until the transaction commits. Only after commit succeeds may they flush.
- Starting a workflow is one effect kind. Example:

```
await deps.effects.call({
  kind: 'workflow.start',
  payload: {
    workflow: 'charge-card',
    input: { customerId: auth.customerId, amount: 4999 },
  },
  preview: 'Charge the customer card for $49.99 via workflow.',
});
```

Provider deps are built from the cycle:

```
type TrainerHandlerDeps = {
  workoutDal: WorkoutDal;
  clientWorkoutDal: ClientWorkoutDal;
  effects: EffectContext;
  logger: Logger;
};

function createTrainerHandlerDeps(args: {
  driver: Driver;
  effects: EffectContext;
  logger: Logger;
}): TrainerHandlerDeps {
  return {
    workoutDal: createWorkoutDal(args.driver),
    clientWorkoutDal: createClientWorkoutDal(args.driver),
    effects: args.effects,
```

```
    logger: args.logger,
  };
}
```

The raw execute context stays separate and is supplied by ai-computer or the HTTP backend:

```
type ExecuteContextFactory = (args: {
  construct: Construct;
  impulse: Impulse;
}) => Promise<Record<string, unknown>>;
```

11.3 Core Implementation (~150 lines)

```
async function runPreview(code: string, config: PreviewConfig):
Promise<PreviewExecRecord> {
  const js = new Bun.Transpiler({ loader: 'ts' }).transformSync(code);

  const forkedFs = config.overlayFs.fork();
  const cycle = await config.transactionRuntime.begin({ mode: 'preview' });
  const stdout: string[] = [];
  const draftResponses: DraftResponse[] = [];
  const effectPreviews: string[] = [];
  const execId = generateExecId();

  const handlers = config.moduleFactory.createHandlers({
    driver: cycle.driver,
    effects: cycle.effects,
    logger: config.logger,
  });

  const context = vm.createContext({
    console: { log: (...a) => stdout.push(a.map(String).join(' ')), warn: ...,
error: ... },
    command: (name, ...args) => config.registry.dispatch(name, args, { fs:
forkedFs }),
    show: createShow(config.registry, stdout),
    draft_response: (r) => draftResponses.push(r),
    TextEncoder, TextDecoder,
    URL, URLSearchParams,
    structuredClone, atob, btoa,
    crypto: globalThis.crypto,
```

```
  });

  const mod = new vm.SourceTextModule(js, { context });
  await mod.link((specifier) => resolveModule(specifier, config, context));
  await mod.evaluate({ timeout: config.timeout ?? 30_000 });

  effectPreviews.push(...cycle.effects.getPreviews());
  await cycle.rollback();

  return {
    execId,
    code,
    stdout: stdout.join('\n'),
    draftResponses,
    effectPreviews,
    createdAt: new Date().toISOString(),
  };
}
```

11.4 TypeScript Stripping

`Bun.Transpiler({ loader: 'ts' }).transformSync(code)` — Bun-native, microsecond speed. For `pm publish`, TypeScript is compiled at publish time. Content-hash cache avoids re-transpiling.

11.5 Timeout and Safety

- **Timeout:** `module.evaluate({ timeout: 30_000 })` — configurable per construct.
- **No `setTimeout` / `setInterval`:** Omitted from globals to prevent long-running or polling code.
- **Threat model:** Code is authored by the LLM, not adversarial users. `vm.createContext` controls API surface. See §22 for full analysis.

11.6 Bun Compatibility

`vm.SourceTextModule`, `vm.SyntheticModule`, and `importModuleDynamically` are all implemented in Bun. No experimental flags. `Bun.Transpiler` is native.

---

12. `sys` Modules

12.1 `sys` — Command Dispatch

A SyntheticModule that exports `command`:

```
const sysModule = new vm.SyntheticModule(['command'], function() {
  this.setExport('command', registry.dispatch.bind(registry));
}, { context });
```

Available as a global (no import needed) and as an import: `import { command } from 'sys';`

12.2 `sys/ai` — Lightweight LLM SDK

Sub-LLM calls inside preview code. The host side manages API calls, budget, and model selection.

Method	Signature	Description
`ai.extract`	`(text: string, zodSchema: ZodType, opts?: { model?: string }) → Promise<T>`	Parse text into structured data
`ai.search`	`(query: string, opts?: { model?: string }) → Promise<SearchResult[]>`	LLM-backed web search
`ai.generate`	`(prompt: string, opts?: { model?: string, budget?: number }) → Promise<string>`	Text generation

Model presets: Developer configures at construct init:

```
ai: {
  models: { fast: 'gpt-4o-mini', thinking: 'o3-mini' },
  budgetPerImpulse: 5000,
}
```

---

13. Node Built-in Proxies

13.1 Tier 1 — Must Have

Module	Implementation	Surface
`node:fs`	SyntheticModule → forked OverlayFs	`readFileSync`, `writeFileSync`, `readFile`, `writeFile`, `existsSync`, `readdirSync`, `mkdirSync`, `statSync`, `unlinkSync`, `renameSync`
`node:path`	SyntheticModule, pure JS	`join`, `resolve`, `dirname`, `basename`, `extname`, `parse`, `format`, `isAbsolute`, `normalize`, `relative`, `sep`
`node:buffer`	SyntheticModule, pure JS polyfill	`Buffer.from`, `Buffer.alloc`, `Buffer.concat`, `toString`, `slice`

13.2 Tier 2 — Should Have

Module	Implementation	Surface
`node:url`	SyntheticModule, pure JS	`URL`, `URLSearchParams`, `parse`, `format`
`node:util`	SyntheticModule, pure JS	`format`, `inspect`, `promisify`, `types.isDate`
`node:crypto` (subset)	SyntheticModule, wrap WebCrypto	`randomUUID`, `createHash`, `randomBytes`
`node:assert`	SyntheticModule, pure JS	`assert`, `strictEqual`, `deepStrictEqual`
`node:querystring`	SyntheticModule, pure JS	`parse`, `stringify`
`node:os` (subset)	SyntheticModule, hardcoded	`EOL`, `platform()`, `tmpdir()`, `homedir()`
`node:events`	SyntheticModule, pure JS	`EventEmitter`

13.3 Tier 3 — Skip for v1

`node:stream`, `node:child_process`, `node:http`, `node:net`, `node:zlib`, `node:worker_threads`. Network/IO access → product bindings.

13.4 `fetch`

Off by default. Opt-in per construct config. Network access goes through product bindings or `sys/ai`.

13.5 Globals in vm Context

- `console` (log, warn, error → captured to stdout)
- `command` (dispatches to registry)
- `show` (command-aware pretty-print + return)
- `draft_response` (propose user-facing response)
- `TextEncoder`, `TextDecoder`
- `URL`, `URLSearchParams`
- `structuredClone`
- `atob`, `btoa`
- `crypto` (WebCrypto)

Not provided: `setTimeout`, `setInterval`, `fetch` (unless opted in).

---

14. Codemode Bindings (Product APIs)

14.1 Defining Bindings

Product functions are defined via `defineCodeBinding` in `.tool.ts` files or via manual `codeFunction()`. The preferred shape is: router handler provider as shared application logic, then a thin code binding wrapper for AI use:

```
// workout-api.tool.ts (build-pack processed)
import { defineCodeBinding } from '@inkibra/ai-flow/codemode';

export const searchExercises = defineCodeBinding(
  async function searchExercises(input: SearchExercisesInput, {
    deps,
  }: {
    deps: {
      handlers: {
        searchExercises: ReturnType<typeof searchExercisesHandlerProvider>;
      };
      executeContext: Record<string, unknown>;
    };
  }): Promise<ExerciseResult[]> {
    return deps.handlers.searchExercises.execute(
      { body: input, pathParams: {}, pathQuery: {} },
      deps.executeContext,
    );
  },
);
```

The handler provider is standard `@inkibra/router` style and receives invocation-scoped deps:

```
export const searchExercisesHandlerProvider = createApiRouteHandlerProvider(
  (deps: {
    exerciseDal: ExerciseDal;
    effects: EffectContext;
    logger: Logger;
  }) =>
    createApiRouteHandler({
      route: searchExercisesRoute,
      handler: async (args, ctx) => {
        const auth = requireTrainerAuth(ctx);
```

```
        const result = await deps.exerciseDal.find(deps.logger, {
          filter: { workspaceId: auth.workspaceId, query: args.body.query },
          limit: 20,
        });
        if (result.isErr()) throw result.error;
        return SerializableResult.toOk(result.value, StatusCode.OK);
      },
    }),
);
```

14.2 Delivery as SyntheticModules

Each module name becomes a `vm.SyntheticModule`. `wrapCodeFunction` validates input, calls `fn` with host context, and returns the result. Bindings should be thin wrappers over router handlers/providers so the same application logic can back both HTTP and AI.

14.3 Host Context

CodeFunction implementations receive `{ deps, fs, ctx, input }`:

- **`deps`**: Invocation-scoped product dependencies (typically handler map, tx-bound DALs, effect context, logger)
- **`fs`**: The OverlayFs instance (forked during preview, real during commit)
- **`ctx`**: Flow state
- **`input`**: Impulse payload

The AI never sees host context. For DB-backed product APIs, the preferred `deps` shape is provider-instantiated router handlers plus a raw execute context.

14.4 Transaction-Bound Deps

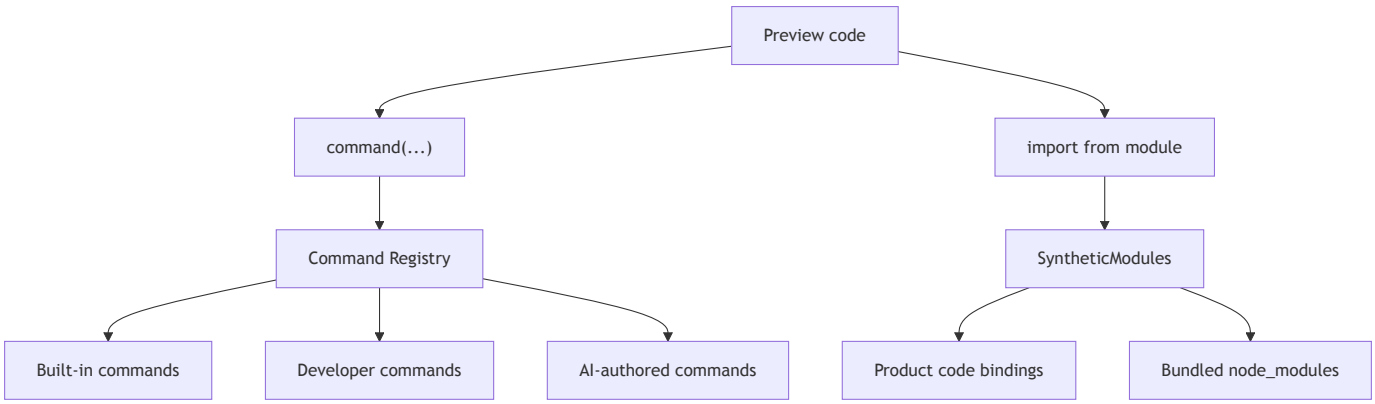
For each top-level server request or AI invocation, the runtime creates:

- a tx-bound `Driver`
- invocation-scoped DAL collections built from that driver
- an invocation-scoped effect context
- provider deps that include those DALs/effects

Bindings do not branch on `isDryRun`. Preview vs commit semantics come from the transaction cycle:

- **Preview:** tx rolls back, effects stay staged
- **Commit / HTTP:** tx commits, then effects flush

14.5 Commands vs Code Bindings



Commands are lightweight program operations (`command('read', ...)`) — string/number args, `render` for `show()`.

Code bindings are typed library imports (`import { getWorkouts } from 'workout-api'`) — typed params, typia-validated, full host context.

Both are available inside `preview_exec`. They serve different roles and are defined differently (`defineCommand` vs `defineCodeBinding`).

---

15. Module Tiers & Resolution Order

```
1. 'sys'              → SyntheticModule (command() global)
2. 'sys/ai'           → SyntheticModule (ai.extract, ai.search, ai.generate)
3. 'node:*'           → SyntheticModule (proxied built-ins, see §13)
4. Binding registry   → SyntheticModule (product bindings, see §14)
5. /usr/lib/node_modules/<name>/
                      → SourceTextModule from VFS (bundled npm: zod, date-fns)
6. /home/user/.local/lib/node_modules/<name>/
                      → SourceTextModule from VFS (ts-pm packages)
7. Not found          → Error
```

---

16. Pre-installed Packages

Developer-bundled npm packages in the system tier:

Package	Size (bundled)	Purpose
`zod`	~50KB	Schema definition for `sys/ai.extract`, AI validation
`date-fns`	~30KB (tree-shaken)	Date formatting and manipulation

Bundled via `esbuild --bundle --platform=browser --format=esm`. The AI cannot install arbitrary npm at runtime.

---

17. ts-pm

17.1 Commands

Sub-command	Args	Description
`pm init <name>`	name	Scaffold `/home/user/scripts/<name>/`
`pm publish <name>`	name	Compile TS→JS, copy to registry (immutable)
`pm add <name>`	name, `--version`	Install. Auto-registers command if default export is a `Command`.
`pm list`	*(none)*	List installed packages
`pm versions <name>`	name	List published versions
`pm remove <name>`	name	Uninstall, unregister command
`pm link <path>`	path	Dev-mode symlink

17.2 Lock Status

Status	Behavior
`locked`	Read-only. System packages.
`draft-propose`	AI can modify. Activation requires approval.
`auto-approve`	AI can modify and publish freely.

17.3 Auto-registration of Commands

When `pm add` installs a package whose default export satisfies the `Command` shape (has `name`, `description`, `args`, `fn`, `render`), it is auto-registered in the command registry. One command per package.

---

18. Intent Hint Registry

18.1 Concept

Intents are **not** a separate runtime system. An intent is a discoverable alias for a concrete command invocation — a TOML entry that maps a natural-language phrase to a command string.

There is no intent server, no dispatch function, no intent handler shape. Intents are command macros that the system can auto-preview.

18.2 Registry Format

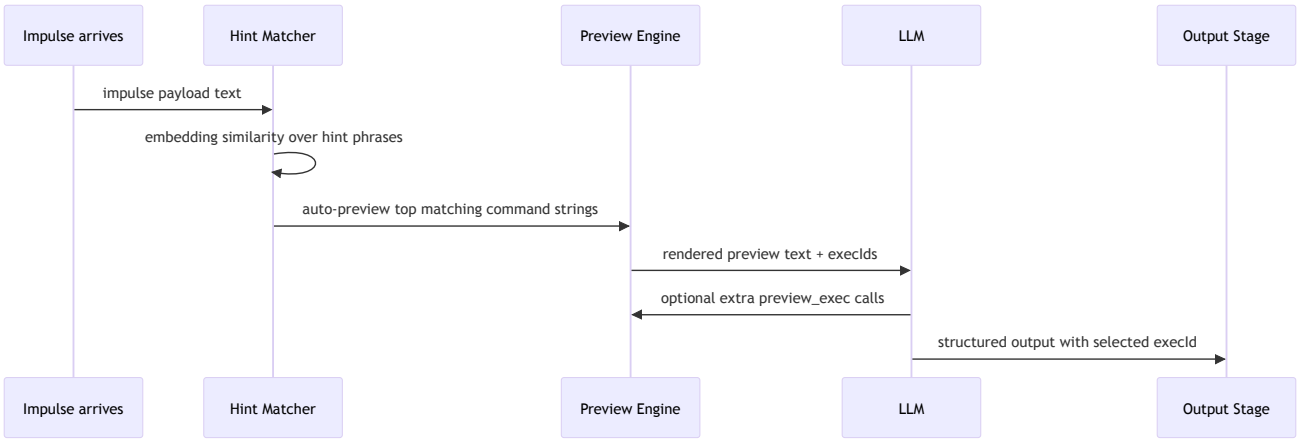
The AI maintains `/home/user/.local/lib/intents/registry.toml`:

```
[[intents]]
intent = "log-last-7d-workouts"
hint = "what have i done this week"
command = "log-workouts --last 7d"

[[intents]]
intent = "check-streak"
hint = "how is my streak"
command = "streak-check"

[[intents]]
intent = "weekly-summary"
hint = "weekly summary"
command = "weekly-report --weeks 1"
```

Each entry has:

- **`intent`**: stable name for the registration
- **`hint`**: natural-language phrase for similarity matching
- **`command`**: full command string (parsed by the command arg parser)

18.3 Intent Hint Lifecycle



1. **Similarity match:** Embedding similarity between impulse payload text and registered `hint` phrases. Top-N matches surfaced.
2. **Auto-preview:** Each matched intent's `command` string is run as a `preview_exec` program. Specifically:

   ```
   show(await command('log-workouts', '--last', '7d'));
   ```

   This produces a `PreviewExecRecord` with an `execId`.
3. **LLM sees previews:** The auto-preview results appear as tool results before the LLM's first turn. The LLM can:
   - Select one of the auto-previewed `execId`s directly
   - Ignore them and use `preview_exec` to build something else
   - Use them as context for its own reasoning

18.4 Adding a New Intent

The AI registers intents by editing the TOML file:

```
// Inside preview_exec during nap:
const registry = await command('read',
'/home/user/.local/lib/intents/registry.toml');
await command('write', '/home/user/.local/lib/intents/registry.toml',
  registry + `\n[[intents]]\nintent = "check-streak"\nhint = "how is my
streak"\ncommand = "streak-check"\n`);
```

The referenced command (`streak-check`) must already exist in the command registry (installed via `pm add`).

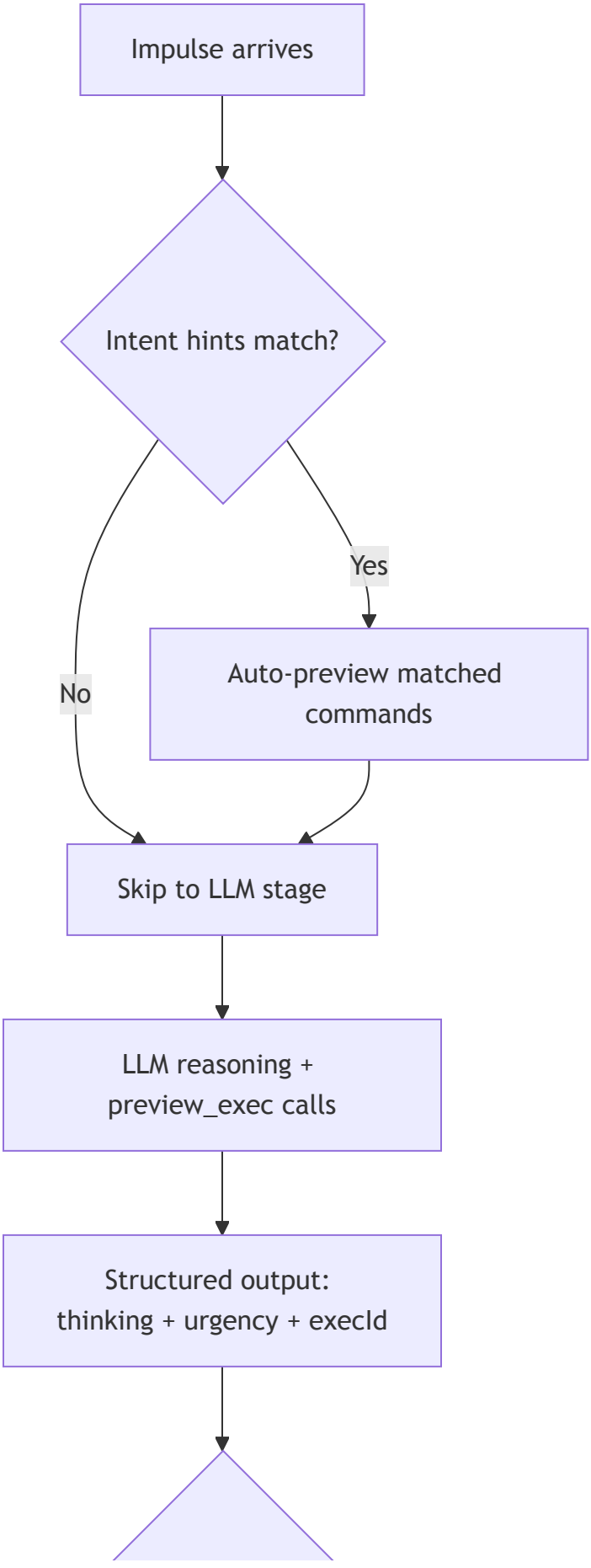
18.5 System vs AI Intents

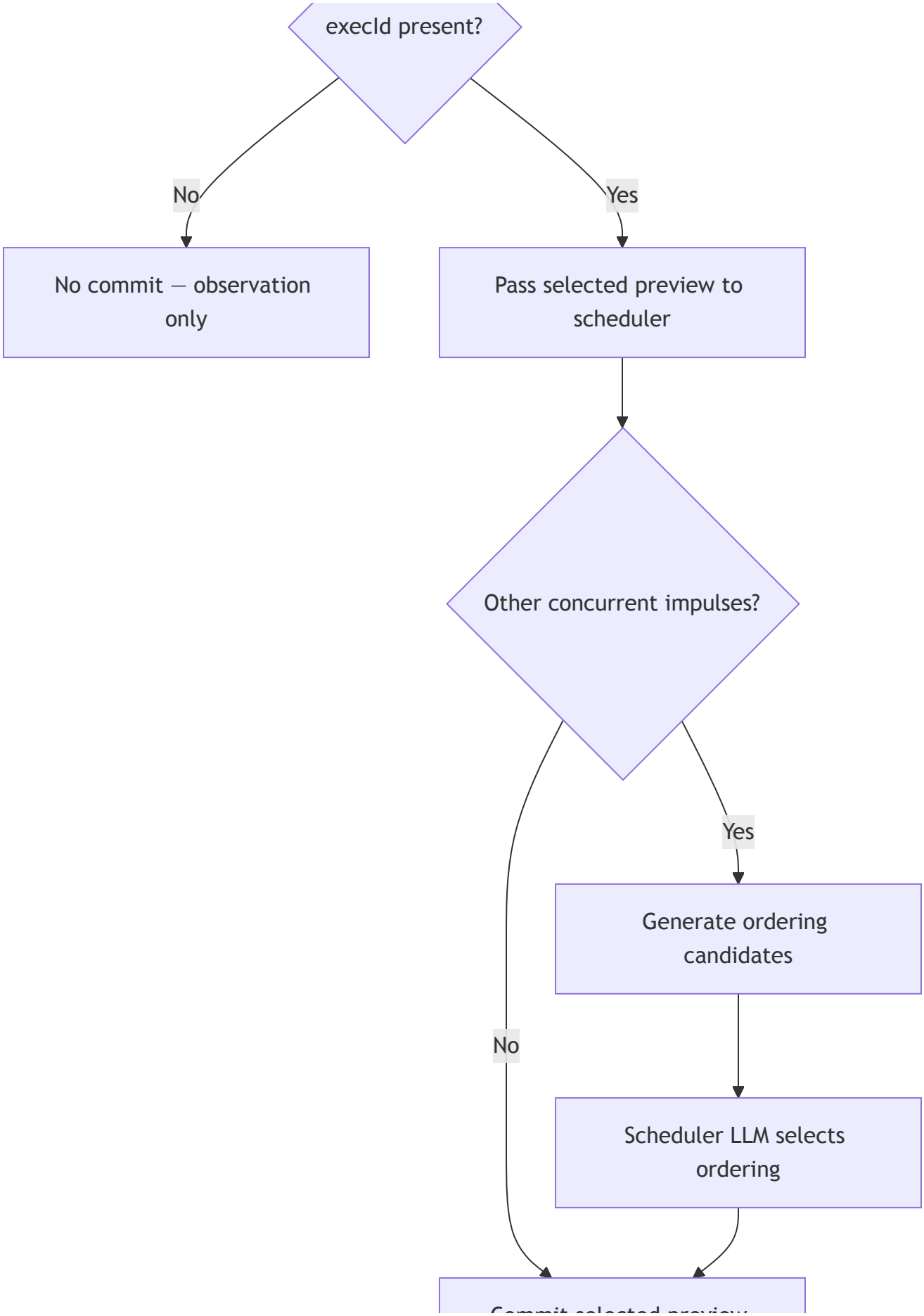
System intents can be pre-registered at `/usr/lib/intents/registry.toml` (read-only to AI). Both registries are scanned by the hint matcher.

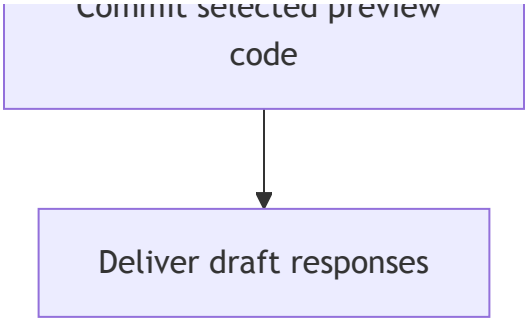
---

19. Impulse Lifecycle

19.1 Overview


Impulse arrives
Intent hints match?
Yes
No
Auto-preview matched commands
Skip to LLM stage
LLM reasoning + preview_exec calls
Structured output: thinking + urgency + execId


execId present?
No
Yes
No commit — observation only
Pass selected preview to scheduler
Other concurrent impulses?
Yes
No
Generate ordering candidates
Scheduler LLM selects ordering



19.2 Stage Breakdown

Pre-LLM: Intent hint matching (ai-construct, no LLM)

- Load intent registries (system + AI)
- Embedding similarity between impulse text and hint phrases
- Top matches → auto-preview the command strings
- Preview results stored in flow context as `PreviewExecRecord`s
- These appear as tool results in the LLM's prompt

LLM turn: Reasoning + preview_exec (ai-flow stage)

- AI has one tool: `preview_exec`
- AI may see auto-previewed results from intent hints
- AI calls `preview_exec` zero or more times to explore
- Each call produces a `PreviewExecRecord` with `execId`

Output stage: Selection (ai-flow structured output)

- AI returns `{ thinking, urgency, execId }`
- `execId` selects one preview record (or `null` for no commit)

Post-output: Scheduler commit (ai-construct)

- Selected preview record passed to scheduler
- Scheduler commits the selected preview by re-running it inside a fresh transaction cycle (see §20)

19.3 How ai-construct Uses ai-flow

```
export async function runImpulse(impulse, vfs, config, deps) {
  // Pre-LLM: intent hint auto-preview
  const hintPreviews = await runIntentHintMatching(impulse, vfs, config);

  // LLM turn
  const computer = createComputer(vfs, config.computer);
  const previewExecTool = createPreviewExecTool({ engine: computer.engine });
  const flow = createImpulseFlow(previewExecTool, config, hintPreviews);
  const result = await flow.start({ impulse, decision: null }).complete();
```

```
  // Post-output: pass to scheduler
  const decision = result.ctx.decision;
  const selectedPreview = decision.execId
    ? result.ctx.previewExecRuns[decision.execId]
    : null;

  await scheduler.submit({
    impulseId: impulse.id,
    thinking: decision.thinking,
    urgency: decision.urgency,
    selectedPreview,
    executeContext: await config.computer.createExecuteContext({
      construct: deps.construct,
      impulse,
    }),
  });
}
```

19.4 `/proc/self/ctx`

Flow state is written to `/proc/self/ctx` as JSON before the LLM turn. The AI can `command('read', '/proc/self/ctx')` to introspect it.

---

20. Scheduler Commit & Ordering Selection

20.1 Single Impulse (Common Case)

When one impulse submits a selected preview and no other impulses are pending for this construct:

1. Retrieve the `PreviewExecRecord` (code + draft responses)
2. Open a fresh transaction cycle and build tx-bound provider deps from it
3. Re-run the code against the **real** OverlayFs and tx-bound DALs
4. Run any final handler-level invariant checks
5. Commit the transaction
6. Flush staged effects only after commit succeeds
7. Deliver draft responses to the response stage

20.2 No Selected Preview

If `execId` is `null`, the impulse was observation-only. Nothing to commit. The impulse is consumed. Draft responses (if any from other mechanisms) still flow through the response stage.

20.3 Concurrent Impulses — Ordering Candidates

When multiple impulses submit selected previews concurrently for the same construct, the framework generates **valid ordering candidates** before invoking the scheduler LLM.

Each ordering candidate is a synthetic preview that:

1. Creates a fresh `OverlayFs.fork()`
2. Opens one preview transaction cycle for DB-backed product APIs
3. Runs the first impulse's selected preview code
4. Then runs the second impulse's selected preview code **on the same fork and same transaction cycle** — so the second code sees the first's VFS writes and uncommitted DB state
5. Captures combined stdout, combined draft responses, combined effect previews, and a combined `execId`

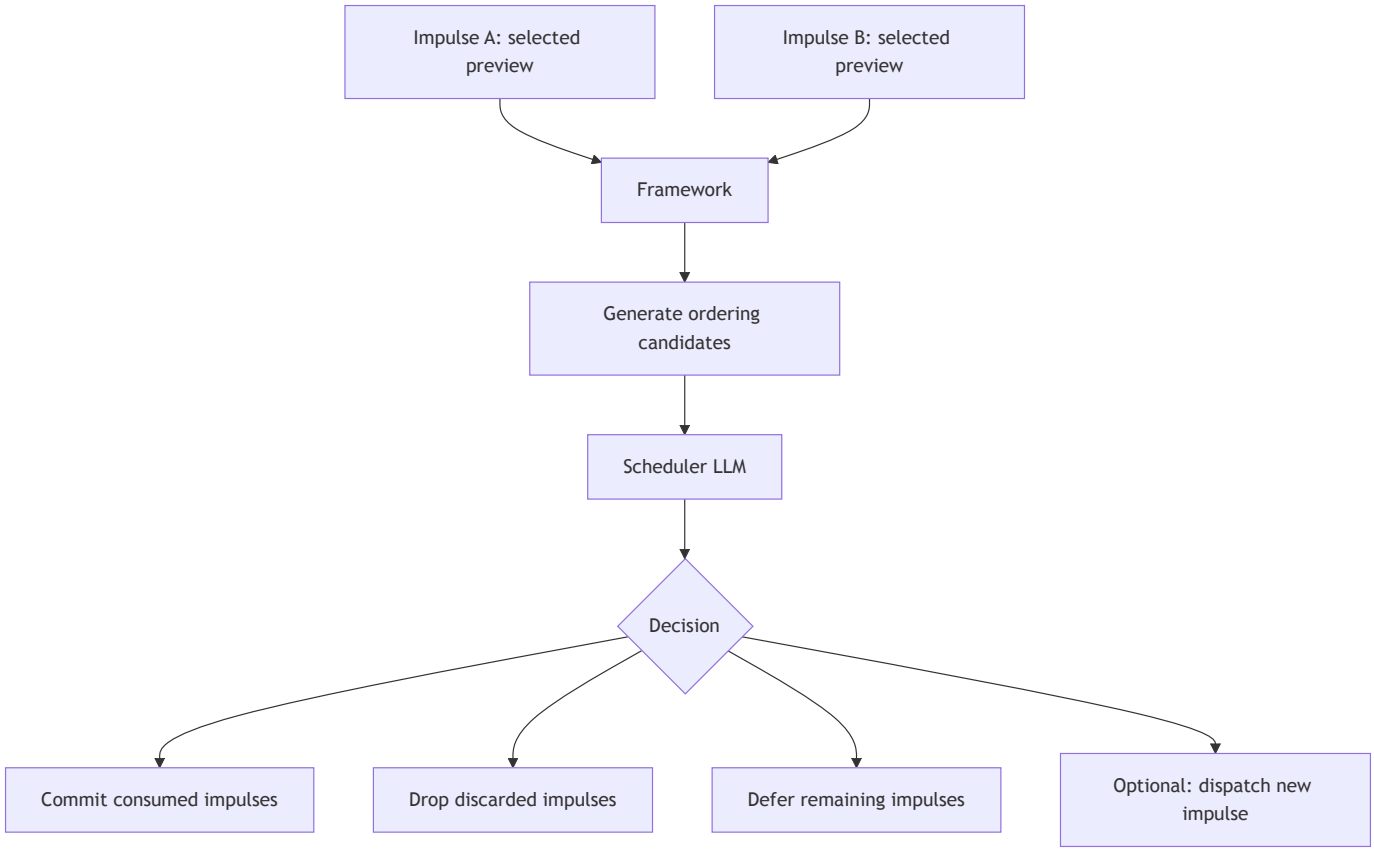
For 2 concurrent impulses, this produces at most 2 candidates:

- Candidate 1: impulse A code → impulse B code
- Candidate 2: impulse B code → impulse A code

Plus the singletons (A alone, B alone) are always available as candidates.

For 3+ concurrent impulses, limit to pairwise orderings or the most promising subset. Full factorial is not feasible beyond 2.

20.4 Scheduler Decision



The scheduler LLM receives:

- Each pending impulse with its thinking, urgency, and preview summary
- The ordering candidate previews with their combined stdout and draft responses
- Current dispatched-impulse budget status

The scheduler does **not** write code. It selects from presented candidates.

Structured output:

```
type SchedulerDecisionOutput = {
  thinking: string;
  consumedImpulses: string[];    // impulse IDs in a valid ordering
  droppedImpulses: string[];     // explicitly discarded
  dispatchImpulse?: {            // omitted from schema if budget exhausted
    reason: string;              // prefiller detects intent from this
  };
};
```

Rules (stated explicitly in the scheduler prompt):

- `consumedImpulses` must exactly match one of the presented valid ordering candidates (framework validates this)
- `droppedImpulses` are permanently discarded — they will not be retried
- Any impulse not in `consumedImpulses` or `droppedImpulses` is **automatically deferred** — it returns to the impulse queue and will be processed in a future scheduler turn with fresh state
- `dispatchImpulse` creates a new impulse with the given `reason` text. The prefiller will detect intent hints from the reason when it fires. This field is **omitted from the schema entirely** when the construct's dispatched-impulse budget is exhausted (prevents the LLM from attempting to dispatch when it cannot)

20.5 Dispatched Impulse Budget

Dispatched impulses are rate-limited per construct to prevent runaway loops.

```
type DispatchBudgetConfig = {
  maxPerHour: number;       // e.g. 10
  maxPerDay: number;        // e.g. 50
};
```

The current budget state is surfaced to the scheduler in its prompt context: "You have dispatched N impulses in the last hour (budget: M)."

When budget is exhausted, the `dispatchImpulse` field is removed from the structured output schema — the scheduler LLM cannot generate it.

20.6 Commit Execution

Once the scheduler selects an ordering:

1. Framework looks up the ordering candidate's synthetic preview
2. Opens one fresh commit transaction cycle
3. Re-runs the combined code against real OverlayFs and tx-bound DALs
4. Runs final invariant checks
5. Commits the transaction
6. Flushes staged effects after commit
7. Draft responses from the real run delivered to response stage

If the real commit fails (e.g., binding call error that didn't occur in preview, invariant failure, or DB commit conflict), the failure is logged and the impulse(s) are deferred for retry in a future scheduler turn. Router-level request/invocation runtimes should expose a generic retryable error such as `TryAgainLater` for commit-time failures.

20.7 Concurrency Model

Impulses for the same construct **can run concurrently** through the LLM reasoning phase (preview_exec calls, structured output). Concurrency issues only arise at commit time. The scheduler serializes commits per construct.

This avoids a global mutex on the impulse lifecycle. LLM calls (the slow part) run in parallel. Only the commit (the fast part) serializes.

---

21. OverlayFs

21.1 Role

`OverlayFs` (`ai-flow/codemode/overlay-fs.ts`, 519 lines) is the construct's virtual filesystem. In-memory overlay on persistent backends. Features: dirty tracking, content-hash dedup, multi-backend path routing, tombstone deletes.

It is the isolation primitive for **filesystem state only**. DB-backed product state uses the router transaction cycle, not OverlayFs.

21.2 Internal State

```
class OverlayFs {
  private memory: Map<string, string>;
  private deleted: Set<string>;
  private dirty: Set<string>;
  private persistedHash: Map<string, string>;
  private persistentPaths: PersistentPathConfig[];
}
```

21.3 `fork()` Method

Critical for the preview model. Every `preview_exec` call runs on a fork:

```
fork(): OverlayFs {
  const forked = new OverlayFs({ persistent: this.persistentPaths });
  forked.memory = new Map(this.memory);
  forked.deleted = new Set(this.deleted);
  forked.dirty = new Set(this.dirty);
  forked.persistedHash = new Map(this.persistedHash);
  forked._cwd = this._cwd;
```

```
  return forked;
}
```

Sub-millisecond at current scale (hundreds to low thousands of files).

21.4 Testing Gaps

`overlay-fs.test.ts` has **44 lines** and **1 test case**. Untested: read/write/delete round-trip, tombstones, multi-backend routing, content-hash dedup, fork() isolation, concurrent flush.

---

22. Security & Threat Model

22.1 Threat Model

Primary assumption: **code running inside preview_exec is authored by the LLM, not by adversarial end users.**

Threat	Mitigation
Sandbox escape	`vm.createContext` isolates globals
Infinite loop	`module.evaluate({ timeout })`
FS access beyond VFS	`node:fs` proxied through forked OverlayFs
Network access	No `fetch` by default; product bindings or `sys/ai`
Cross-agent data	OverlayFs per-agent + CodeFunction auth per userId
Preview side effects	Transaction cycle rolls back and effect context never flushes

22.2 Data Isolation Layers

```
Layer 1: OverlayFs per-agent (no shared mutable state between constructs)
Layer 2: Handler/provider auth (execute context + provider deps)
Layer 3: Scheduler commit serialization (per-construct, no concurrent commits)
```

22.3 Arbitrary User Code

Delegate to external MCP services (E2B, Modal) via a CodeFunction binding.

---

23. Deployment Scenarios & Scaling

23.1 Scenario A — ToneTempo (Single Product, Many Users)

One product. Many users with independent constructs. Shared system tier.

- Peak concurrent preview_exec at 1,000 agents: ~20-50
- OverlayFs memory per agent: ~100KB typical
- Concurrency via async event loop interleaving
- fork() overhead: sub-ms per preview

23.2 Scenario B — Zerbly (ai-construct as a Service)

Multi-tenant. Per-tenant isolation of bindings, modules, and configuration.

23.3 Scaling Lever

Postgres backing for OverlayFs — cold files evicted from memory, re-loaded on demand.

---

24. Migration Plan

24.1 What Changes

Current	New
`createVfsTool` (bash tool)	Merged into `createPreviewExecTool`
`createExecuteTool` (execute tool)	Merged into `createPreviewExecTool`
`ContextShell` (2,739 lines)	Command handlers via `defineCommand` (~800 lines)
`executor.ts` (609 lines, vm.createContext + memfs)	Preview engine (~200 lines, vm.SourceTextModule + fork)
Named globals (`await fetchUser(...)`)	ESM imports (`import { getWorkouts } from 'workout-api'`)
`ctx` object mutation	File writes to `/home/user/context/state/`
Two+ tools (bash + execute)	One tool (`preview_exec`)
Structured output: `{ response, intentDispatch }`	Structured output: `{ thinking, urgency, execId }`
Intent server + dispatch + prefiller	Intent hint TOML + similarity + auto-preview

Current	New
Self-dispatch impulses	Removed for v1
`isDryRun` / query-cache DB preview semantics	Router transaction cycle + rollback
Inline external handling in bindings	Staged effects flushed after commit

24.2 Phase 1a — Command Registry + Built-in Handlers (1–1.5 weeks)

#	Task	Difficulty
1	`CommandRegistry` class + `defineCommand` interface + arg parser	Medium
2	File operation handlers: read, write, edit, list, search, find, stat, remove, mkdir	Easy
3	Port open/close/opened/pin/unpin/status (context management)	Hard
4	Port cron (7 sub-commands)	Medium
5	`show()` implementation with command tag + render dispatch	Easy
6	Port context-shell tests	Medium
7	Expand overlay-fs tests	Medium

24.3 Phase 1a-parallel — ts-pm (1 week)

#	Task	Difficulty
1	pm init, publish (Bun.Transpiler), add, list, versions, remove, link	Medium
2	Lock status enforcement	Easy
3	Auto-registration: detect `Command` default export on `pm add`	Medium

24.4 Phase 1b — Preview Engine (1.5–2 weeks)

#	Task	Difficulty
1	`runPreview()` — vm.SourceTextModule + Bun.Transpiler + OverlayFs.fork()	Medium
2	Linker function — module resolution (7-tier)	Medium
3	`sys` SyntheticModule (command dispatch)	Easy
4	Product binding SyntheticModules (CodeFunction → SyntheticModule)	Medium
5	Node proxy SyntheticModules (Tier 1 + 2)	Medium

#	Task	Difficulty
6	`createPreviewExecTool` with dual render/context storage	Medium
7	`draft_response()` capture in preview records	Easy
8	Preview record storage in ai-flow context	Easy
9	Port executor tests to preview model	Medium

24.5 Phase 2 — Package Extraction + Impulse Integration (2–3 weeks)

#	Task	Difficulty
1	`@inkibra/ai-sandbox-computer` package creation	Easy
2	`createComputer(overlayFs, config)` assembly function	Easy
3	`sys/ai` SyntheticModule (extract, search, generate)	Medium
4	AI-installed command loading from VFS	Medium
5	OverlayFs `fork()` method	Easy
6	Impulse structured output: `{ thinking, urgency, execId }`	Medium
7	Post-output validation (execId → preview record lookup)	Easy
8	Scheduler commit: re-run selected code inside fresh transaction cycle	Medium

24.5a Parallel Cross-Package Runtime Work

#	Task	Difficulty
1	`@inkibra/router`: add transaction-cycle and effect-context interfaces	Hard
2	`@inkibra/dal-connection`: implement transaction-cycle for supported drivers	Hard
3	`@inkibra/denzel-bun`: request lifecycle wrapper that opens tx + binds effects	Medium
4	`@inkibra/workflow`: effect backend for workflow-start intents	Medium
5	ai-computer: per-invocation execute-context callback and tx-bound handler instantiation	Medium
6	Preview-safe vs preview-unsafe route metadata	Medium

24.6 Phase 3 — Intent Hints + Ordering Selection (3–4 weeks)

#	Task	Difficulty
1	Intent hint TOML parser	Easy
2	Embedding similarity matcher	Medium
3	Auto-preview of matched commands at impulse start	Medium
4	Inject auto-preview results into LLM prompt context	Medium
5	Concurrent impulse detection in scheduler	Medium
6	Ordering candidate generation + scheduler selection stage	Hard
7	Response delivery for selected ordering candidate	Medium
8	Nap prompt context (impulse patterns, existing intents)	Easy

24.7 What Does Not Change

- OverlayFs core and its backing store options
- OpenAI streaming orchestration in ai-flow
- Impulse pool, construct lifecycle, runtime manager
- Nap and hypno flow stage structure (only prompt context changes)
- `createAiFlow`, `createAiOutputStage`, `createAiTextStage` APIs
- `.tool.ts` build-pack pipeline (output becomes SyntheticModules)
- No production migration concerns — breaking changes are acceptable

---

25. Resolved Decisions

- **One tool: `preview_exec`.** The AI explores and drafts with preview code. Structured output selects one `execId`. The scheduler commits it.
- **Preview/commit separation.** Nothing commits during the LLM turn. All previews run on forked OverlayFs and a transaction cycle that always rolls back. The scheduler is the sole commit authority.
- **Commands, not a shell.** No bash, no pipes. `command('read', path)` is a function call. Non-shell names prevent shell syntax attempts.
- **Commands vs code bindings.** `command()` is for lightweight program operations. ESM imports are for typed product API libraries. Different roles, different definition interfaces (`defineCommand` vs `defineCodeBinding`).
- **Router handlers/providers are the shared application layer.** Server and AI both execute the same provider-instantiated handlers. Provider `deps` receive tx-bound DALs and staged effects; handler `ctx` remains auth/request context.

- **`draft_response()` in code.** Response proposals are written programmatically after the AI knows computed results. `urgency` stays in structured output.
- **Intent hints, not intent dispatch.** Intents are TOML entries pointing at command strings. Auto-previewed at impulse start. No intent server, no dispatch function, no separate handler shape.
- **No self-dispatch for v1.** Replaced by cron scheduling of command invocations.
- **Same-thread execution.** `vm.SourceTextModule` on the main thread. No Workers, no SAB. Workers are a v2 optimization.
- **No `setTimeout` / `setInterval`.** Prevent long-running or polling code.
- **`fetch` off by default.** Network → product bindings or `sys/ai`.
- **`show()` uses command's `render`.** Each command defines how its result is pretty-printed. `show()` returns the raw value.
- **`preview_exec` is a text-input custom tool.** The model writes raw TypeScript as the tool input, not a JSON object. ai-flow needs to support custom / text-input tools alongside JSON-schema function tools. OpenAI Responses API supports this. If a provider does not, ai-flow falls back to wrapping as `{ code: string }`.
- **One top-level transaction cycle per invocation.** DB-backed preview state comes from a tx-bound driver, not `isDryRun` branches. Preview rolls back; commit reruns inside a fresh cycle and commits.
- **Effects are fire-and-forget.** Handlers can only enqueue effects and get preview text back. Real delivery happens only after commit succeeds. Starting a workflow is one effect kind.
- **Ordering selection, not code merge.** For concurrent impulses, the framework generates synthetic ordering previews (A→B, B→A). The scheduler LLM selects from these — it **never writes code**. Scheduler actions: consume (valid ordering), drop, defer (automatic for unmentioned impulses), or dispatch a new impulse (budget-limited, reason-only — prefiller handles intent detection). The scheduler does not merge, rewrite, or generate code.
- **Preview-safe route subset for AI.** Routes that mutate Redis/cache/session state or other non-transactional side stores are preview-unsafe in v1 and are not exposed through ai-computer.

---

26. Open / To Be Designed

Preview Model

- **Preview record retention:** How long are preview records kept in flow context? Cleared after the output stage? After scheduler commit?
- **Auto-render last expression:** If preview code has no stdout and no `show()`, should the last expression auto-render?
- **Module caching:** Can compiled SourceTextModules be cached across preview calls within one impulse?
- **Effect preview text shape:** Is `preview?: string` enough, or should effects return richer structured preview metadata?

Router / Runtime Integration

- **Transaction-cycle API shape:** Exact router-level interface for open / commit / rollback / effect flush.
- **Effect context API:** `call()` only, or named typed effect methods plus a generic fallback?
- **Execute-context mapping:** Exact callback contract between ai-construct's per-impulse auth/context and ai-computer's raw router execute context.
- **Preview-safe metadata:** How routes/handlers declare preview-safe vs preview-unsafe behavior.

Intent Hints

- **Embedding model choice:** Which embedding model for similarity matching. Cold-start for new intents with no usage history.
- **Auto-preview budget:** How many intent hints can be auto-previewed per impulse? Default top-N.
- **Command-string parse failures:** What happens if a registered intent's `command` string fails to parse against the command's arg schema?
- **Preview failure handling:** If an auto-previewed command errors, is the error shown to the LLM as a failed preview, or is the hint silently dropped?

Scheduler Ordering & Commit

- **Ordering candidate limit:** For 3+ concurrent impulses, which orderings to preview? Full factorial is too expensive. Pairwise? Most-urgent-first?
- **Commit failure handling:** If real commit fails (binding error that didn't occur in preview), impulse is deferred. But how many retries before drop?
- **Ordering candidate preview budget:** Each candidate costs one fork + one run. At what point do we skip ordering previews and just serialize?
- **Scheduler prompt design:** How much VFS state snapshot to include? Just file list? Include content of key state files?
- **Dispatch budget tuning:** Default `maxPerHour` / `maxPerDay` values. Should budget be configurable per impulse profile?

sys/ai

- **Budget enforcement:** Per-impulse or per-preview-call?
- **Streaming:** `ai.generate` streaming or always full result?
- **Model presets:** Default preset set.

ai-flow Custom Tool Support

- **Exact wire format:** Confirm OpenAI Responses API custom-tool request shape, streaming event types, and SDK support for text-input tools.
- **Provider fallback:** If a provider does not support custom tools, ai-flow wraps as `{ code: string }` function tool. Verify this works transparently.

- **Event handling:** Confirm that streaming tool-call events for custom tools are handled correctly in ai-flow's OpenAI streaming orchestration.

Scaling (v2)

- **Worker threads:** When CPU-bound preview becomes a bottleneck.
- **OverlayFs memory eviction:** LRU for cold file eviction to postgres.
- **Bun Secure Mode:** PR #25911. Worth tracking.